

Note 5.5 Indefinite Integrals and the Substitution Method

**Relationship among Indefinite Integral, Antiderivatives and Definite Integral

- **Indefinite Integral and Antiderivatives:**

The indefinite integral of the function f with respect to x is the set of all antiderivatives of f . That is,

$$\int f(x) dx = F(x) + C$$

where $F(x)$ is any antiderivative of f and C is any arbitrary constant.

- **Antiderivatives and Definite Integral:**

It is stated in the [fundamental theorem](#).

- **Indefinite Integral and Definite Integral:**

A definite integral $\int_a^b f(x) dx$ is a number while an indefinite integral $\int f(x) dx$ is a function plus an arbitrary constant C .

Substitution: Running the Chain Rule Backwards

- **The Substitution:**

As with differentials, when computing integrals we have

$$du = \frac{du}{dx} dx$$

- **Theorem 6--The Substitution Rule:**

If $u = g(x)$ is a differentiable function whose range is an interval I , and f is continuous on I , then

$$\int f(g(x))g'(x) dx = \int f(u) du$$

Proof:

We can suppose that F is an antiderivative of f . By [the Chain Rule](#),

$$\frac{d}{dx} F(g(x)) = F'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(x)$$

thus when we make the substitution $u = g(x)$, then

$$\begin{aligned}\int f(g(x))g'(x) dx &= \int \frac{d}{dx} F(g(x)) dx \\ &= F(g(x)) + C \\ &= F(u) + C \\ &= \int F'(u) du \\ &= \int f(u) du\end{aligned}$$

- **Substitution Method:**

- Substitute $u = g(x)$ and $du = \frac{du}{dx} \cdot dx = g'(x)$ to obtain the integral

$$\int f(u) du$$

- Integrate with respect to u .
- Replace u by $g(x)$ in the result.

The Integrals of $\sin^2 x$ and $\cos^2 x$

- **Strategy:**

Use the formulas about trigonometric functions and the substitution rule.